



ABBOTSLEIGH

Student's Name:

Student Number:

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Teacher's Name:

2025

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- Diagrams are not to scale unless indicated
- Ensure your HSC candidate number is on each booklet
- For questions in Section I, answer on the sheet provided
- For questions in Section II,
 - show all necessary working
 - start a new booklet for each question

Total marks: **Section I – 10 marks** (pages 3–7)

100

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8–16)

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Year 12 Mathematics Extension 2 outcomes

MEX12-1 understands and uses different representations of numbers and functions to model, prove results and find solutions to problems in a variety of contexts

MEX12-2 chooses appropriate strategies to construct arguments and proofs in both practical and abstract settings

MEX12-3 uses vectors to model and solve problems in two and three dimensions

MEX12-4 uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems

MEX12-5 applies techniques of integration to structured and unstructured problems

MEX12-6 uses mechanics to model and solve practical problems

MEX12-7 applies various mathematical techniques and concepts to model and solve structured, unstructured and multi-step problems

MEX12-8 communicates and justifies abstract ideas and relationships using appropriate language, notation and logical argument

Section I

10 Marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1. What is the value of i^{2025} ?

A. 1

B. -1

C. i

D. $-i$

2. A cubic polynomial $P(z)$ has all real coefficients.

$z = 3$ and $z = 2 + i$ are two roots of $P(z) = 0$.

Which equation describes $P(z)$?

A. $P(z) = z^3 - 7z^2 - 17z + 15$

B. $P(z) = z^3 - 7z^2 + 17z - 15$

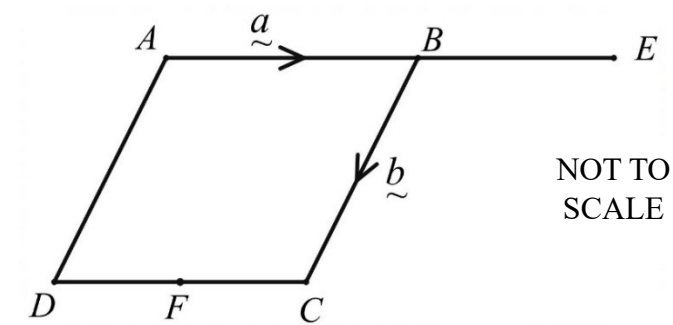
C. $P(z) = z^3 + 7z^2 - 35z + 15$

D. $P(z) = z^3 + 7z^2 + 35z - 15$

3. Given that $z = \frac{\sqrt{3}+i}{1+i}$, which of the following is equal to z^5 ?

- A. $2\sqrt{2}e^{\frac{\pi}{12}i}$
 B. $4\sqrt{2}e^{\frac{\pi}{12}i}$
 C. $2\sqrt{2}e^{\frac{19\pi}{12}i}$
 D. $4\sqrt{2}e^{\frac{19\pi}{12}i}$

4. $ABCD$ is a rhombus. It is given that $\overrightarrow{AB} = \underline{a}$, and $\overrightarrow{BC} = \underline{b}$.



$\overrightarrow{BE}$ is an extension of $\overrightarrow{AB}$ such that $\overrightarrow{AB} : \overrightarrow{BE} = 4 : 3$.

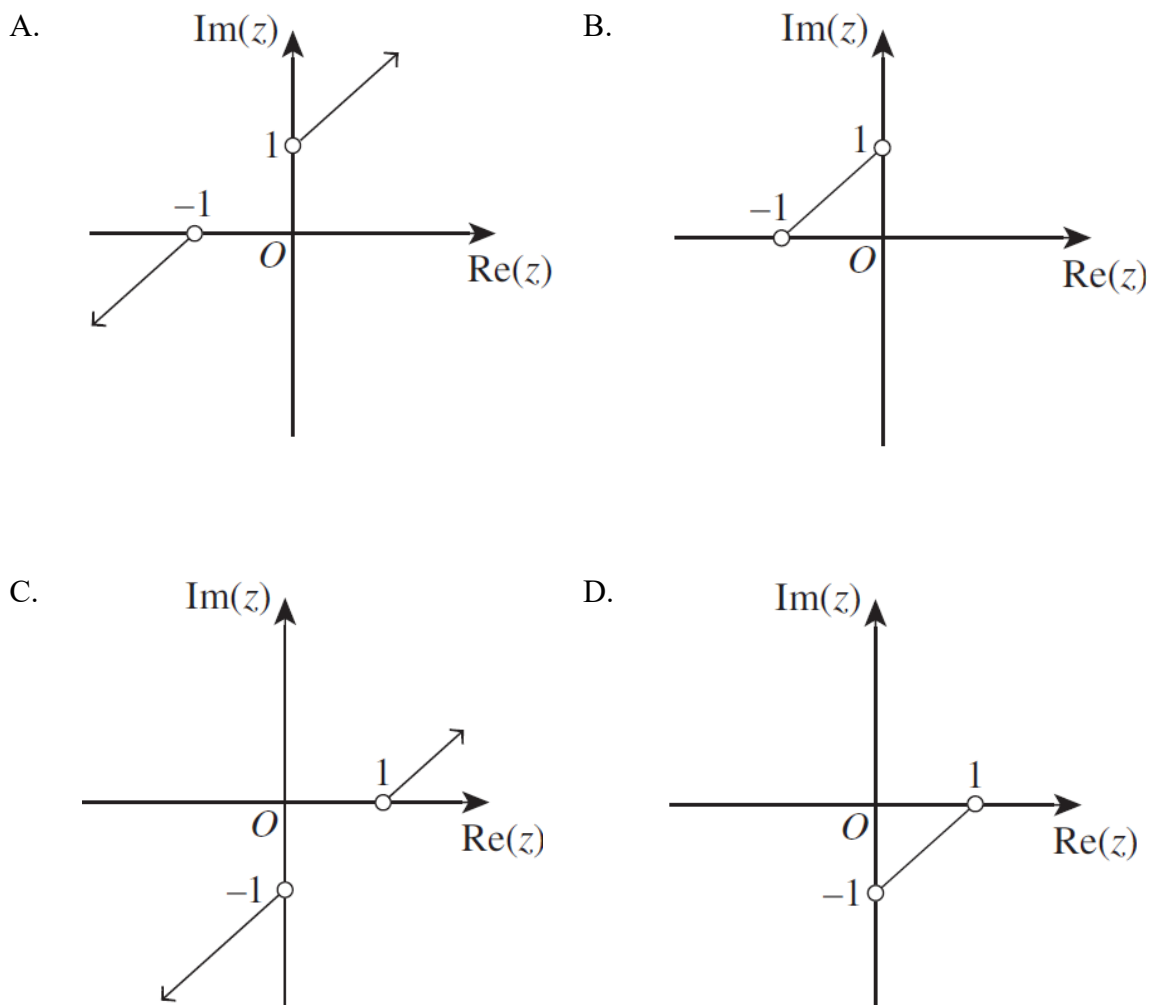
The point F is the midpoint of CD .

Which expression gives the vector $\overrightarrow{FE}$ in terms of $\underline{a}$ and $\underline{b}$?

- A. $\frac{11}{6}\underline{a} + \underline{b}$
 B. $\frac{5}{4}\underline{a} + \underline{b}$
 C. $\frac{11}{6}\underline{a} - \underline{b}$
 D. $\frac{5}{4}\underline{a} - \underline{b}$

5. Which of the following Argand diagrams represents the solutions to the equation:

$$\text{Arg}(z - i) = \text{Arg}(z + 1)?$$



6. Which of the statements is logically equivalent to the one shown opposite.

- A. If you may enter, then you are authorised.
- B. If you are authorised, then you may enter.
- C. If you are not authorised, then you may enter.
- D. If you may not enter, then you are not authorised.



7. A particle of **unit** mass is projected vertically upwards from the ground with initial velocity of u m/s.

The particle is affected by a gravitational force, g , and a resistive force equivalent to $\frac{1}{2}kv^2$,

where the particle's velocity is v m/s. Let k be the constant of proportionality and x be the displacement, in metres, from the ground after t seconds.

Which of the following is the maximum height, H , of the particle?

A.
$$H = \int_u^0 \frac{v}{g + \frac{kv^2}{2}} dv$$

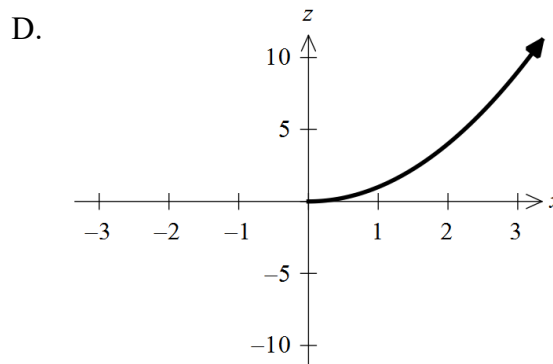
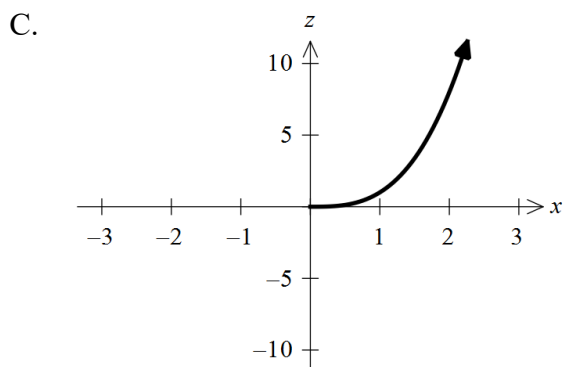
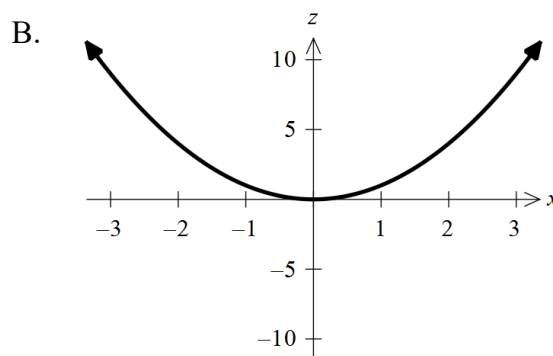
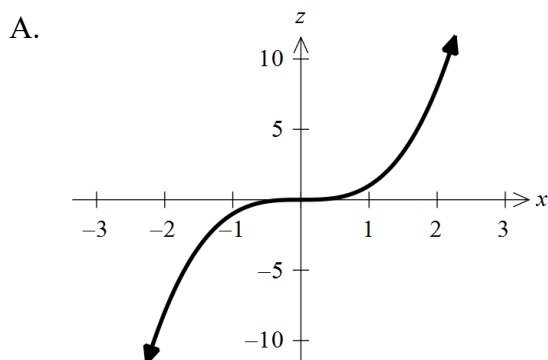
B.
$$H = \int_0^u \frac{v}{g + \frac{kv^2}{2}} dv$$

C.
$$H = \int_u^0 \frac{v}{g - \frac{kv^2}{2}} dv$$

D.
$$H = \int_0^u \frac{v}{g - \frac{kv^2}{2}} dv$$

8. The curve $r(t) = (t^2, t^4, t^6)$ is defined for all real values of t .

Which of the following graphs best represents its **projection onto the xz -plane**?



9. A car journey is split into two parts.

During the first part the average speed is V km/h.

In the second part, the average speed is U km/h.

Kimie concludes that the average speed for the whole journey is $\frac{U+V}{2}$ km/h.

Which of the following is NOT a counterexample to this conclusion?

- A. The car initially travels 100 km at 20 km/h, and then travels 100 km at 40 km/h.
- B. The car initially travels 120 km at 60 km/h and 80 km at 40 km/h.
- C. The car initially travels for one hour at 100 km/h and then two hours at 50 km/h.
- D. The car initially travels 60 km in two hours and then travels 40 km in three hours.

10. Consider the definite integral, $I = \int_0^{\pi} x \sin x \, dx$.

Using the substitution $x = \pi - y$, what is the value of I ?

- A. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin y \, dy$
- B. $\pi \int_0^{\pi} \sin y \, dy$
- C. $\frac{\pi}{2} \int_0^{\pi} \sin y \, dy$
- D. $\int_0^{\pi} \sin y \, dy$

End of Section I

Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a **SEPARATE** writing booklet. Extra writing booklets are available.

In Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z_1 = -1 - 5i$ and $z_2 = 2 - i$.

(i) Calculate $i(z_1 - z_2)$. 1

(ii) Write $\frac{a + ib}{z_2}$ in the form $\omega = x + iy$, in terms of a and b . 2

(b) Consider the statement about integers 2

If $k^2 - 14k$ is odd, then k is odd.

By considering the contrapositive prove that the statement is true.

(c) Let $P \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be any point that is equidistant from $Q \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$ and $R \begin{pmatrix} -7 \\ 1 \\ 0 \end{pmatrix}$.

(i) Show that the equation of all such points P is $18x - 8y + 8z = -21$. 2

(ii) Give a geometric description of P . 1

(d) Solve $z^2 + (5 + 5i)z + (6 + 15i) = 0$. Give your answer in the form $z = x + iy$. 3

(e) Use partial fractions to find $\int_{-2}^1 \frac{1 + 7x}{(x - 2)(x + 3)} dx$. 4

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to prove that if $S_1 = a$ and $S_n - S_{n-1} = ar^{n-1}$, $n \geq 2$, 3

$$\text{then } S_n = \frac{a(1-r^n)}{1-r}, \quad n \geq 1.$$

- (b) Consider the complex number $z = 1 + i\sqrt{3}$.

- (i) Write z in mod-arg form. 1

- (ii) Show $\sqrt{z^3} = 2\sqrt{2}i$. 1

- (iii) Find all values of n for which z^n is purely imaginary. 1

- (c) (i) Find the Cartesian equation for the line l that passes through the point $\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$ 2

$$\text{and is parallel to the line with equation } \vec{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}.$$

- (ii) Determine where the line l meets the x - z plane. 2

- (d) Prove that $\log_2 9$ is irrational. 2

- (e) If ω is a complex solution of $z^3 = 1$, simplify the expression $(1 + \omega)^2 + (1 + \omega^2)^2$. 2

- (f) Write the negation of the following statement $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}^+ : xy = 1$. 1

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^1 \log_e(x^2 + 1) dx$. 4

(b) Consider the spheres S_1 and S_2 2

$$S_1 : x^2 - 2x + y^2 + 4y + z^2 - 4z = 0$$

$$S_2 : x^2 + y^2 + z^2 + 10x + 2z + 10 = 0.$$

Explain why S_1 and S_2 intersect at a single point. (Do NOT find this point.)

(c) Consider the complex number $z = \cos \theta + i \sin \theta$.

(i) Show that $z^n - \frac{1}{z^n} = 2i \sin n\theta$. 1

(ii) Hence, or otherwise, find the constants a , b and c such that 3

$$\sin^5 \theta = a \sin 5\theta + b \sin 3\theta + c \sin \theta.$$

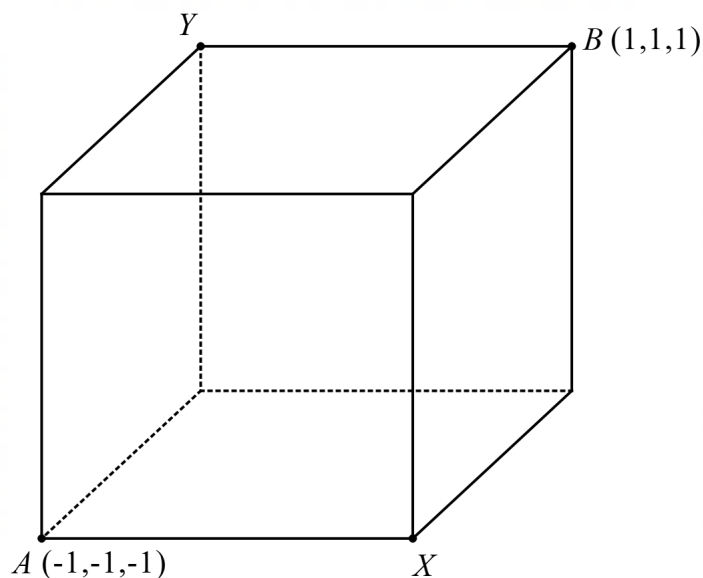
(iii) Hence evaluate $16 \int_0^{\frac{\pi}{2}} \sin^5 \theta d\theta$. 1

Question 13 continues over page

Question 13 (continued)

- (d) A cube has a pair of opposite vertices $A(-1, -1, -1)$ and $B(1, 1, 1)$. 2

Another pair of opposite vertices are X and Y , as displayed in the diagram below.



If α is the acute angle between AB and XY , show that $\sin 2\alpha = \frac{2\sqrt{8}}{9}$.

- (e) Three positive number a , b and c satisfy the following conditions 2

$$a \geq b \geq c \text{ and } a + b + c \leq 1.$$

By considering $(a + b + c)^2$ or otherwise, prove that $a^2 + 3b^2 + 5c^2 \leq 1$.

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Use the substitution of $u = \sin x - x \cos x$, evaluate **3**

$$\int_{\frac{\pi}{2}}^{\pi} \frac{x}{1 - x \cot x} dx.$$

- (b) Consider the following four equations, where $a > 0$ and $b > 0$. **3**

$$D = \frac{2ab}{a+b}$$

$$U = \sqrt{ab}$$

$$B = \frac{a+b}{2}$$

$$O = \sqrt{\frac{a^2 + b^2}{2}}$$

Show that $D \leq U \leq B \leq O$.

- (c) The triangle ABC in the complex plane is formed from the points $A(\alpha)$, $B(\beta)$ and $C(\gamma)$. **3**

The complex numbers α , β and γ satisfy these two conditions:

$$\frac{\gamma - \alpha}{\beta - \alpha} = 1 - i \quad \text{and} \quad \alpha + \beta + \gamma = 0.$$

Calculate $|\alpha| : |\beta|$.

Question 14 continues over page

Question 14 (continued)

- (d) Two magpies, A and B , are observed flying. 3

Their initial respective positions are:

$(\hat{i} - 2\hat{j} + 4\hat{k})$ metres and $(-2\hat{i} + a\hat{j} + 6\hat{k})$ metres, where a is a constant.

They are travelling with constant velocities and these are respectively:

$(2\hat{i} + 3\hat{j} + 6\hat{k})$ m/s and $(3\hat{i} + 12\hat{j} + 4\hat{k})$ m/s.

Determine the constant a , if the distance between the two magpies is least after 5 seconds.

- (e) By considering $(1+i)(4-i)^2$ prove that $\tan^{-1}\left(\frac{7}{23}\right) + 2\tan^{-1}\left(\frac{1}{4}\right) = \frac{\pi}{4}$. 3

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) A body of **unit** mass falls vertically from rest, under gravity, from a height of 100 metres above the ground through a resisting medium. The resistance to its motion is $\frac{v^2}{100}$, where v is the speed of the body (in m/s) after falling a distance x metres.

(i) Show that the acceleration $\ddot{x}$ of the body is given by $\ddot{x} = g - \frac{v^2}{100}$. 1

(ii) Determine the terminal velocity V of the body. 1

(iii) Show that the velocity v of the body, after it has fallen a vertical distance x , is given by 3

$$v^2 = V^2 \left(1 - e^{-\frac{x}{50}} \right).$$

(iv) Determine the distance fallen when the velocity of the body first reaches $\frac{V}{2}$. 2

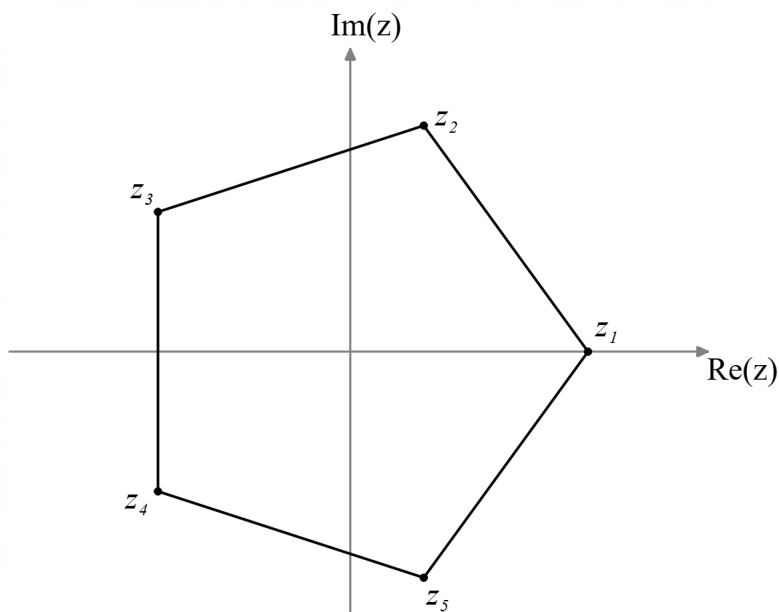
(b) Let $S_n = \sum_{p=1}^n \tan^{-1} \left(\frac{1}{2p^2} \right)$. 4

Prove by induction that $\tan(S_n) = \frac{n}{n+1}$.

Question 15 continues over page

Question 15 (continued)

- (c) The solutions of $z^5 = 1$ can be joined to form a regular pentagon in the complex plane, as shown below.



- (i) For $n \geq 3$, show that the perimeter of the regular n -gon formed by the solutions of $z^n = 1$ is $P_n = 2n \sin\left(\frac{\pi}{n}\right)$. 2

- (ii) It is known that as $x \rightarrow 0$ then $\frac{\sin x}{x} \rightarrow 1$. (DO NOT prove this.) 2

Using part (i), show that as $n \rightarrow \infty$, $P_n \rightarrow 2\pi$.

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

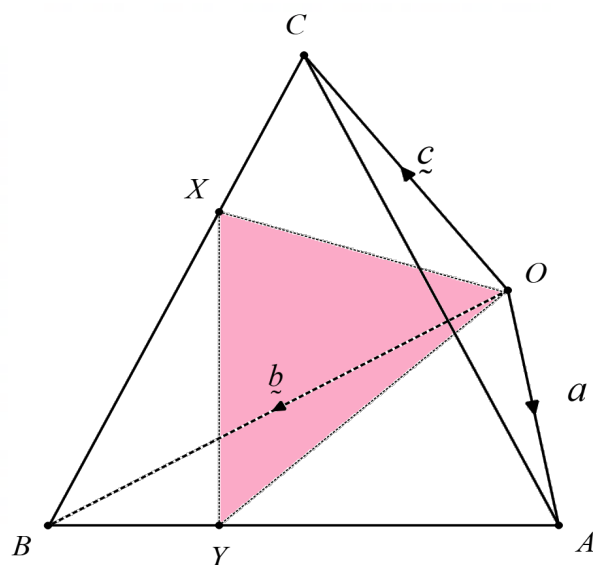
(a) Let $I_n = \int_0^{\frac{1}{4}} \frac{x^n}{e^x} dx$ where $n = 1, 2, 3, \dots$.

(i) Calculate I_1 . 2

(ii) Show that $kI_k = I_{k+1} - I_k + \frac{\left(\frac{1}{4}\right)^{k+1}}{e^4}$. 3

(iii) Deduce that $\sum_{k=1}^n kI_k = I_{n+1} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^{n+1}}{12e^4}$. 3

(b) The diagram shows a regular tetrahedron $OABC$ with sides of length 1.



Let X and Y be points on the edges such that $|\overrightarrow{BX}| : |\overrightarrow{XC}| = |\overrightarrow{AY}| : |\overrightarrow{YB}| = x : 1 - x$, where $0 < x < 1$.

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, and $\overrightarrow{OC} = \underline{c}$.

(i) Show that $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{c} = \frac{1}{2}$. 1

(ii) Show that $|\overrightarrow{OY}| = \sqrt{1 - x + x^2}$. 2

(iii) Determine the range of values of $\angle YOX$. 4

End of Paper

2025 Abbotsleigh Extension 2 Trial Solutions

1. What is the value of i^{2025} ?

- A. 1
- B. -1
- C. i
- D. $-i$

As

$$i^4 = 1$$

$$i^{2024} = 1$$

$$i^{2025} = i$$

Answer is C

2. A cubic polynomial $P(z)$ has all real coefficients.

$z = 3$ and $z = 2 + i$ are two roots of $P(z) = 0$.

Which equation describes $P(z)$?

- A. $P(z) = z^3 - 7z^2 - 17z + 15$
- B. $P(z) = z^3 - 7z^2 + 17z - 15$
- C. $P(z) = z^3 + 7z^2 - 35z + 15$
- D. $P(z) = z^3 + 7z^2 + 35z - 15$

Roots are $z = 3, 2 + i$, and $2 - i$ (complex conjugate).

Sum of roots,

$$-\frac{b}{a} = 3 + (2 + i) + (2 - i) = 7$$

$$b = -7$$

So, answer is either A or B.

Product of roots,

$$-\frac{d}{a} = 3(2 + i)(2 - i) = 15$$

$$d = -15$$

So, the answer is B.

3. Given that $z = \frac{\sqrt{3} + i}{1 + i}$, which of the following is equal to z^5 ?

A. $2\sqrt{2}e^{\frac{\pi}{12}i}$

B. $4\sqrt{2}e^{\frac{\pi}{12}i}$

C. $2\sqrt{2}e^{\frac{19\pi}{12}i}$

D. $4\sqrt{2}e^{\frac{19\pi}{12}i}$

$$\sqrt{3} + i = 2\text{cis}\left(\frac{\pi}{6}\right)$$

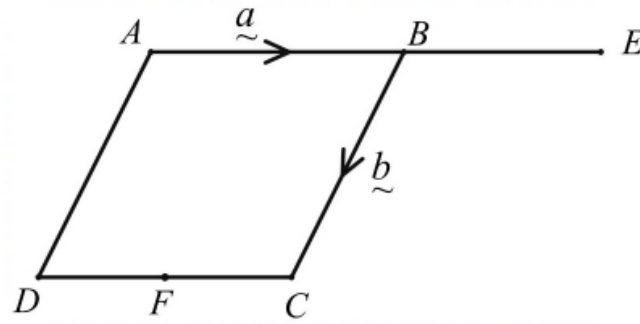
$$1 + i = \sqrt{2}\text{cis}\left(\frac{\pi}{4}\right)$$

$$z = \frac{2\text{cis}\left(\frac{\pi}{6}\right)}{\sqrt{2}\text{cis}\left(\frac{\pi}{4}\right)} = \frac{2}{\sqrt{2}}\text{cis}\left(\frac{\pi}{6} - \frac{\pi}{4}\right) = \sqrt{2}\text{cis}\left(-\frac{\pi}{12}\right)$$

$$z^5 = \sqrt{2}^5 \text{cis}\left(-\frac{5\pi}{12}\right) = 4\sqrt{2}\text{cis}\left(-\frac{5\pi}{12}\right) = 4\sqrt{2}\text{cis}\left(\frac{19\pi}{12}\right)$$

So, the answer is D.

4. $ABCD$ is a rhombus. $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{BC} = \underline{b}$.



$\overrightarrow{BE}$ is an extension of $\overrightarrow{AB}$, such that $\overline{AB} : \overline{BE} = 4 : 3$.

The point F is the midpoint of CD .

Which expression gives the vector $\overrightarrow{FE}$ in terms of $\underline{a}$ and $\underline{b}$?

- A. $\frac{11}{6}\underline{a} + \underline{b}$
- B. $\frac{5}{4}\underline{a} + \underline{b}$
- C. $\frac{11}{6}\underline{a} - \underline{b}$
- D. $\frac{5}{4}\underline{a} - \underline{b}$

$$\overrightarrow{FE} = \overrightarrow{FC} + \overrightarrow{CB} + \overrightarrow{BE}$$

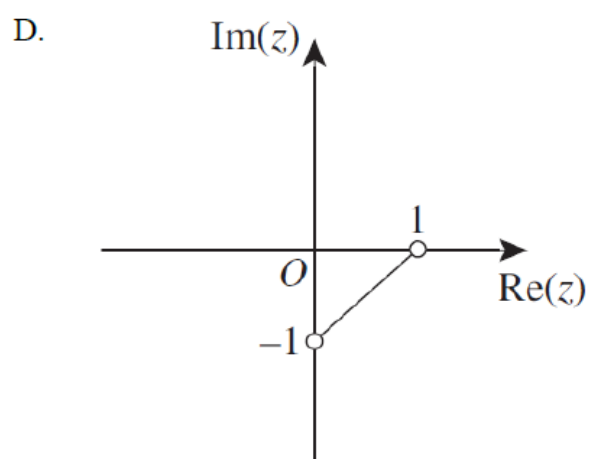
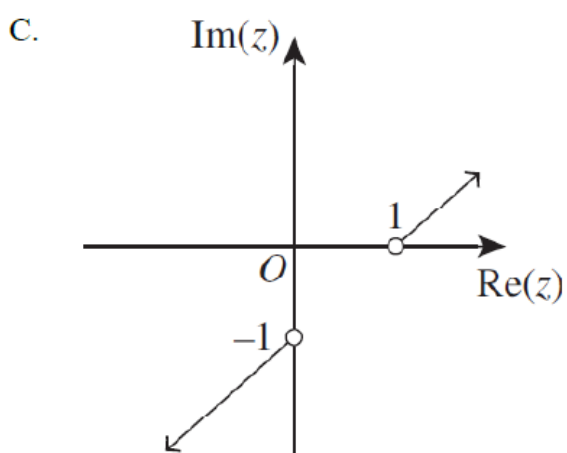
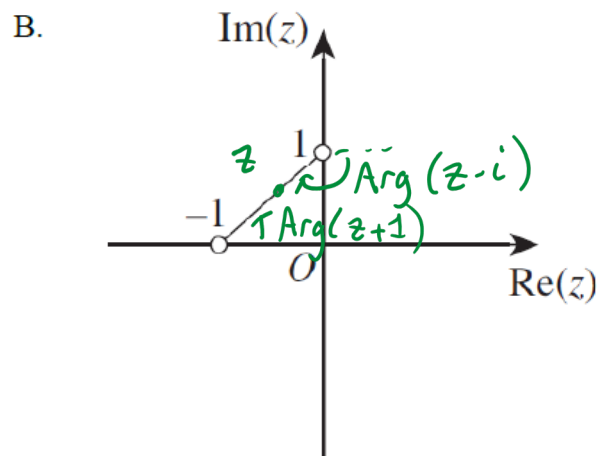
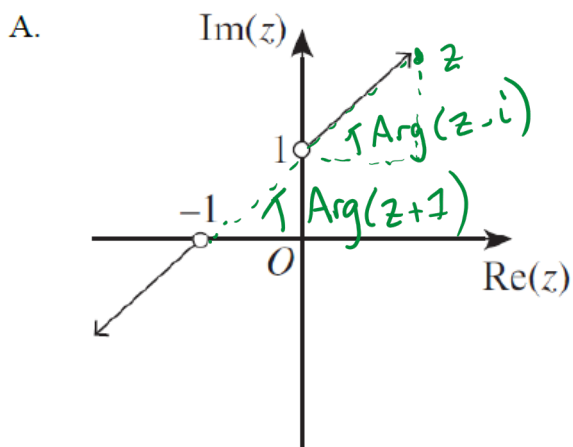
$$\text{As } AB:BE = \frac{4}{3} \rightarrow BE = \frac{3}{4}AB = \frac{3}{4}\underline{a}$$

$$\overrightarrow{FE} = \frac{1}{2}\underline{a} - \underline{b} + \frac{3}{4}\underline{a}$$

$$\overrightarrow{FE} = \frac{5}{4}\underline{a} - \underline{b}$$

So, the answer is D.

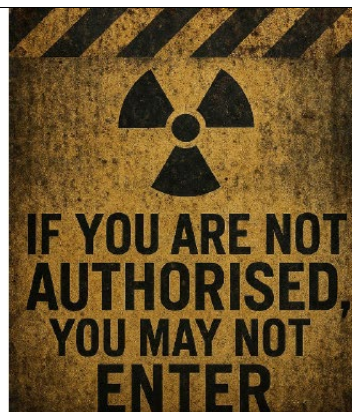
5. Which of the following Argand diagrams represents the solutions to the equation:
 $\text{Arg}(z - i) = \text{Arg}(z + 1)$?



The argument formed from z and the point $(0, 1)$ is equal to the argument formed from z and the point $(-1, 0)$.

From the diagram above you can see that B results in the arguments being supplementary rather than equal – hence the answer is A.

6. Which of the statements is logically equivalent to the one shown opposite.
- A. If you may enter, then you are authorised
- B. If you are authorised, then you may enter
- C. If you are not authorised, then you may enter
- D. If you may not enter, then you are not authorised



(Not authorised) $\rightarrow$ (May not enter)

Is equivalent to its contrapositive,

(May enter) $\rightarrow$ (Authorised)

So, the answer is A.

7. A particle of **unit** mass is projected vertically upwards from the ground with initial velocity of u m/s. The particle is affected by a gravitational force, g , and a resistive force equivalent to $\frac{1}{2}kv^2$, where the particle's velocity is v m/s. Let k be the constant of proportionality and x be the displacement, in metres, from the ground after t seconds. Which of the following is the maximum height, H , of the particle?

A. $H = \int_u^0 \frac{v}{g + \frac{kv^2}{2}} dv$

B. $H = \int_0^u \frac{v}{g + \frac{kv^2}{2}} dv$

C. $H = \int_u^0 \frac{v}{g - \frac{kv^2}{2}} dv$

D. $H = \int_0^u \frac{v}{g - \frac{kv^2}{2}} dv$

Equation of motion being projected upwards, so gravity and resistance are negative. Finding x in terms of v .

$$a = -g - \frac{1}{2}kv^2$$

$$v \frac{dv}{dx} = -\left(g + \frac{1}{2}kv^2\right)$$

$$-\frac{v}{g + \frac{kv^2}{2}} dv = dx$$

Journey to max height is $x: 0 \rightarrow H$, and $v: u \rightarrow 0$.

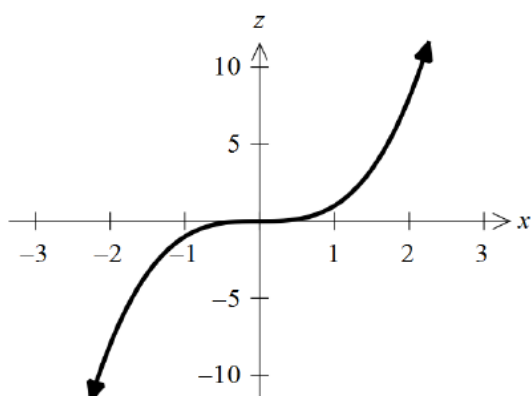
$$-\int_u^0 \frac{v}{g + \frac{kv^2}{2}} dv = \int_0^H dx$$

Flipping the sign of the integral with the negative,
The answer is B.

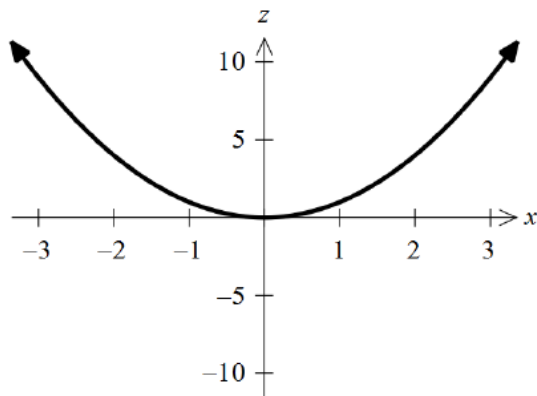
8. The curve $r(t) = (t^2, t^4, t^6)$ is defined for all real values of t .

Which of the following graphs best represents its **projection onto the xz -plane**?

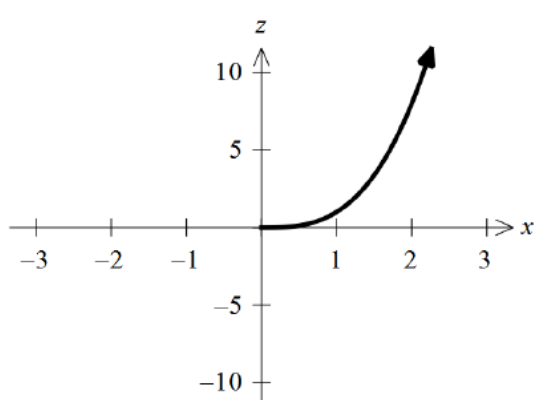
A.



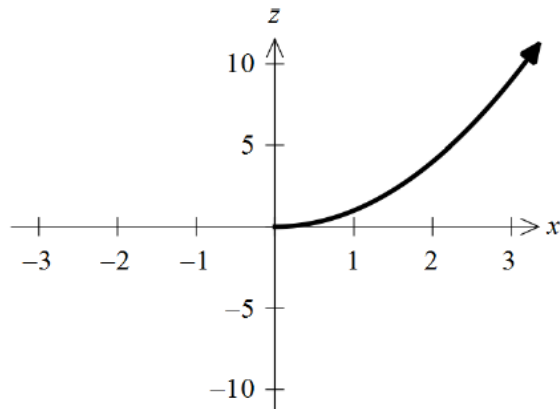
B.



C.



D.



Projection onto the xz -plane implies that $y = 0$.

$$x = t^2, \text{ and } z = t^6$$

$$z = (t^2)^3 = x^3$$

But, as $x = t^2$, and t is a real value $x \geq 0$.

So, **the answer is C**, as this BEST shows a cubic, compared to D.

9. A car journey is split into two parts.

During the first part the average speed is V km/h.

In the second part, the average speed is U km/h.

Kimie concludes that the average speed for the whole journey is $\frac{U+V}{2}$ km/h.

Which of the following is NOT a counterexample to this conclusion?

- A. The car initially travels 100 km at 20 km/h, and then travels 100 km at 40 km/h.
- B. The car initially travels 120 km at 60 km/h and 80 km at 40 km/h.
- C. The car initially travels for one hour at 100 km/h and then two hours at 50 km/h.
- D. The car initially travels 60 km in two hours and then travels 40 km in three hours.

Going through each option:

A:

$$\frac{U + V}{2} = 30 \text{ km/h}$$

Total distance, 200km.

$$\text{Total time, } \frac{100}{20} + \frac{100}{40} = 7.5 \text{ hours}$$

Average speed, $\frac{200}{7.5} = 26.6 \text{ km/h}$ Not, 30km/h, so this is a counterexample.

B:

$$\frac{60 + 40}{2} = 50 \text{ km/h}$$

Total distance, $120 + 80 = 200 \text{ km}$.

$$\text{Total time } \frac{120}{60} + \frac{80}{40} = 4 \text{ hours}$$

Average speed, $\frac{200}{4} = 50 \text{ km/h}$ So, this is NOT a counterexample!

The answer is B.

(Alternatively, thinking about it – if they travel for the same time in both legs then the average speed matches the equation).

10. Consider the definite integral, $I = \int_0^{\pi} x \sin x \, dx$.

Using the substitution $x = \pi - y$, what is the value of I ?

- A. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin y \, dy$
- B. $\pi \int_0^{\pi} \sin y \, dy$
- C. $\frac{\pi}{2} \int_0^{\pi} \sin y \, dy$
- D. $\int_0^{\pi} \sin y \, dy$

$$x = \pi - y$$

$$\frac{dx}{dy} = -1$$

$$dx = -dy$$

$$x = \pi = \pi - y$$

$$y = 0$$

$$x = 0 = \pi - y$$

$$y = \pi$$

$$I = \int_0^{\pi} x \sin x \, dx = \int_{\pi}^0 (\pi - y) \sin(\pi - y) \times -dy$$

Flipping the sign first, then applying obtuse angle fact, then expanding brackets.

$$I = \int_0^{\pi} (\pi - y) \sin(\pi - y) \, dy$$

$$I = \int_0^{\pi} (\pi - y) \sin(y) \, dy$$

$$I = \int_0^{\pi} \pi \sin(y) \, dy - \int_0^{\pi} y \sin(y) \, dy$$

Note, that second integral is what we started with!

$$I = \int_0^{\pi} \pi \sin(y) \, dy - I$$

$$I = \frac{\pi}{2} \int_0^{\pi} \sin(y) \, dy$$

Answer is C.

Question 11

(a) Let $z_1 = -1 - 5i$ and $z_2 = 2 - i$.

(i) Calculate $i(z_1 - z_2)$.

1

(ii) Write $\frac{a+ib}{z_2}$ in the form $\omega = x + iy$, in terms of a and b .

2

$$i(z_1 - z_2) = i[(-1 - 5i) - (2 - i)]$$

$$i(z_1 - z_2) = i(-3 - 4i) = 4 - 3i$$

$$\frac{a+ib}{z_2} = \frac{a+ib}{2-i}$$

$$\frac{a+ib}{2-i} \times \frac{2+i}{2+i} = \frac{2a+ai+2bi-b}{2^2+1^2}$$

$$\frac{2a-b}{5} + i\frac{a+2b}{5}$$

(b) Consider the statement about integers

If $k^2 - 14k$ is odd, then k is odd.

Prove by contrapositive that the statement is true.

2

Contrapositive,

$$(k \text{ is even}) \rightarrow (k^2 - 14k \text{ is even})$$

Assume $k = 2M$, where M is an integer

$$k^2 - 14k = (2M)^2 - 14 \times 2M$$

$$k^2 - 14k = 2(2M^2 - 14M)$$

Hence, $k^2 - 14k$ is even as M is an integer.

(c) Let $P \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be any point that is equidistant from $Q \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$ and $R \begin{pmatrix} -7 \\ 1 \\ 0 \end{pmatrix}$.

(i) Show that the equation of all such points P is $18x - 8y + 8z = -21$. 2

(ii) Give a geometric description of P . 1

$$|QP| = |RP|$$

$$\sqrt{(x-2)^2 + (y+3)^2 + (z-4)^2} = \sqrt{(x+7)^2 + (y-1)^2 + (z)^2}$$

Squaring and expanding...

$$x^2 - 4x + 4 + y^2 + 6y + 9 + z^2 - 8z + 16 = x^2 + 14x + 49 + y^2 - 2y + 1 + z^2$$

Simplifying...

$$29 - 50 = 18x - 8y + 8z$$

$$18x - 8y + 8z = -21$$

As required.

P is a plane that bisects PQ and is perpendicular to PQ. (Plane was needed)

(d) Solve $z^2 + (5+5i)z + (6+15i) = 0$. Give your answer in the form $z = x + iy$. 3

Quadratic formula.

$$z = \frac{-(5+5i) \pm \sqrt{(5+5i)^2 - 4(6+15i)}}{2}$$

$$z = \frac{-5-5i \pm \sqrt{25+50i-25-24-60i}}{2} = \frac{-5-5i \pm \sqrt{-24-10i}}{2}$$

Need to find $\sqrt{-24-10i} = a + ib$

$$-24-10i = a^2 + 2abi - b^2$$

$$a^2 - b^2 = -24, \quad \text{and } -10 = 2ab$$

$$ab = -5$$

By observation, $a = -1, b = 5$ so...

$$z = \frac{-5-5i \pm (-1+5i)}{2}$$

$$z = \frac{-6}{2} = -3$$

Or

$$z = \frac{-4-10i}{2} = -2-5i$$

(e) Find $\int_{-2}^1 \frac{1+7x}{(x-2)(x+3)} dx$.

Partial fractions...

$$\frac{1+7x}{(x-2)(x+3)} \equiv \frac{A}{x-2} + \frac{B}{x+3}$$

$$1+7x \equiv A(x+3) + B(x-2)$$

$$1+7x \equiv 3A - 2B + (A+B)x$$

$$1 = 3A - 2B, \text{ and } 7 = A + B$$

$$B = 7 - A$$

$$1 = 3A - 2(7 - A)$$

$$1 = 5A - 14$$

$$A = 3 \rightarrow B = 4$$

$$\int_{-2}^1 \frac{1+7x}{(x-2)(x+3)} dx = \int_{-2}^1 \left(\frac{3}{x-2} + \frac{4}{x+3} \right) dx$$

$$\left[3 \ln|x-2| + 4 \ln|x+3| \right]_{-2}^1$$

$$(3 \ln 1 + 4 \ln 4) - (3 \ln 4 + 4 \ln 1) = \ln 4$$

Question 12

- (a) Use mathematical induction to prove that if $S_1 = a$ and $S_n - S_{n-1} = ar^{n-1}$, $n \geq 2$,
then $S_n = \frac{a(1-r^n)}{1-r}$, $n \geq 1$.

3

Base case: $n = 1$

$$S_1 = a \text{ (given)}$$

Formula,

$$S_1 = \frac{a(1-r^1)}{1-r} = a \text{ (as required)}$$

As required.

Assume: $n = k$

$$S_k = \frac{a(1-r^k)}{1-r}$$

Prove: $n = k + 1$

$$S_{k+1} - S_k = ar^{k+1-1} \text{ (by definition)}$$

$$S_{k+1} = ar^k + \frac{a(1-r^k)}{1-r} \text{ (by assumption)}$$

$$S_{k+1} = \frac{ar^k(1-r) + a(1-r^k)}{1-r}$$

$$S_{k+1} = \frac{ar^k - ar^{k+1} + a - ar^k}{1-r}$$

$$S_{k+1} = \frac{a(1-r^{k+1})}{1-r}$$

As required.

(b) Consider the complex number $z = 1 + i\sqrt{3}$.

(i) Write z in mod-arg form.

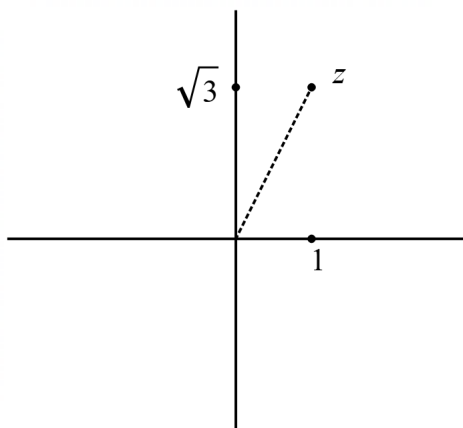
1

(ii) Show $\sqrt{z^3} = 2\sqrt{2}i$

1

(iii) Find all values of n for which z^n is purely imaginary.

1



$$\text{Arg}(z) = \arctan \frac{\sqrt{3}}{1} = \frac{\pi}{3}$$

$$\text{Mod}(z) = \sqrt{1^2 + \sqrt{3}^2} = 2$$

$$z = 2 \text{ cis } \frac{\pi}{3}$$

$$z^3 = 2^3 \text{ cis } \frac{3\pi}{3} = 8 \cos \pi + 8 \sin \pi = -8$$

$$\sqrt{z^3} = \sqrt{-8} = 2\sqrt{2}i$$

$$z^n = 2^n \text{ cis } \frac{\pi n}{3}$$

Purely imaginary means $\cos\left(\frac{\pi n}{3}\right) = 0$

$$\frac{\pi}{3}n = \frac{\pi}{2} + \pi k$$

Where k is an integer.

$$n = \frac{3}{2} + 3k = \frac{6k + 3}{2}$$

(c) (i) Find the Cartesian equation for the line l that passes through the point $\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$

and is parallel to the line with equation $\vec{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$. 2

(ii) Determine where the line l meets the xz plane. 2

$$l = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$$

$$x = 1 + 5\mu$$

$$\mu = \frac{x-1}{5}$$

$$y = -2 + 2\mu$$

$$\mu = \frac{y+2}{2}$$

$$z = 4 - \mu$$

$$\mu = 4 - z$$

$$\frac{x-1}{5} = \frac{y+2}{2} = 4 - z$$

Line meets xz plane when $y = 0$.

$$\mu = \frac{2}{2} = 1$$

$$x = 1 + 5 \times 1 = 6$$

$$z = 4 - 1 = 3$$

$$(6, 0, 3)$$

(d) Prove that $\log_2 9$ is irrational.

2

Assume rational.

$$\log_2 9 = \frac{a}{b}, \text{ where } a \text{ and } b \text{ are integers with no common factors } > 1$$

$$2^{\frac{a}{b}} = 9$$

$$2^a = 9^b$$

But, as a and b are both integers, then $2^a = \text{EVEN}$ and $9^b = \text{ODD}$.

So, this cannot be true.

Therefore $\log_2 9$ is not rational.

(e) If ω is a complex solution of $z^3 = 1$, simplify the expression $(1 + \omega)^2 + (1 + \omega^2)^2$.

2

$$z^3 = 1$$

$$z^3 - 1 = 0$$

From factorizing, and knowing $z = 1$ is a root.

$$(z - 1)(z^2 + z + 1) = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$(\omega^2 + \omega + 1) = 0$$

$$\omega^2 + \omega = -1, \quad (\omega + 1) = -\omega^2 \text{ and } (\omega^2 + 1) = -\omega$$

$$(1 + \omega)^2 + (1 + \omega^2)^2 = (-\omega^2)^2 + (-\omega)^2$$

$$(1 + \omega)^2 + (1 + \omega^2)^2 = \omega^4 + \omega^2$$

$$(1 + \omega)^2 + (1 + \omega^2)^2 = \omega^3 \times \omega + \omega^2 = \omega + \omega^2 = -1$$

NOTE: Expanding the expressions early will reduce to the correct result, but this is not recommended as the powers could be larger (e.g. $(1 + \omega^2)^{10}$, good luck expanding that!)

(f) Write the negation of the following statement $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}^+ : xy = 1$.

1

$$\neg \exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}^+ : xy = 1$$

Is equivalent to...

$$\forall x \in \mathbb{R}^+, \neg \forall y \in \mathbb{R}^+ : xy = 1$$

Is equivalent to

$$\forall x \in \mathbb{R}^+, \exists y \in \mathbb{R}^+ : \neg(xy = 1)$$

$$\forall x \in \mathbb{R}^+, \exists y \in \mathbb{R}^+ : xy \neq 1$$

For all real numbers, there exists another number such that their product is not one.

Question 13

(a) Evaluate $\int_0^1 \log_e(x^2 + 1) dx$.

4

By parts.

$$u = \ln(x^2 + 1)$$

$$v = x$$

$$\frac{du}{dx} = \frac{2x}{x^2 + 1}$$

$$\frac{dv}{dx} = 1$$

$$\int_0^1 \ln(x^2 + 1) dx = \left(\ln(x^2 + 1) \times x \right)_0^1 - \int_0^1 \frac{2x}{x^2 + 1} \times x dx$$

$$\int_0^1 \ln(x^2 + 1) dx = (\ln 2 - \ln 1) - \int_1^e \frac{2x^2 + 2 - 2}{x^2 + 1} dx$$

$$\int_0^1 \ln(x^2 + 1) dx = \ln 2 - \int_0^1 \left(2 - \frac{2}{x^2 + 1} \right) dx$$

$$\int_0^1 \ln(x^2 + 1) dx = \ln 2 - \left(2x - 2 \tan^{-1} x \right)_0^1$$

$$\int_0^1 \ln(x^2 + 1) dx = \ln 2 - [2 - 2 \tan^{-1} 1 - (0 - 2 \tan^{-1} 0)]$$

$$\int_0^1 \ln(x^2 + 1) dx = \ln 2 - 2 + \frac{\pi}{2}$$

(b) Consider the spheres S_1 and S_2 , where

$$S_1 : x^2 - 2x + y^2 + 4y + z^2 - 4z = 0$$

$$S_2 : x^2 + y^2 + z^2 + 10x + 2z + 10 = 0$$

Explain why S_1 and S_2 intersect at a single point. (Do NOT find this point.)

2

$$S_1 : (x - 1)^2 + (y + 2)^2 + (z - 2)^2 = 1 + 4 + 4 = 9$$

Sphere with centre $(1, -2, 2)$ and radius 3 units.

$$S_2 : (x + 5)^2 + (y)^2 + (z + 1)^2 = -10 + 25 + 1 = 16$$

Sphere with centre $(-5, 0, -1)$ and radius 4 units.

- **Distance between the centres?**

$$\sqrt{6^2 + 2^2 + 3^2} = 7$$

Equivalent to the sum of the radii – hence they will touch once.

(c) Consider the complex number $z = \cos \theta + i \sin \theta$.

(i) Show that $z^n - \frac{1}{z^n} = 2i \sin n\theta$. 1

(ii) Hence, or otherwise, find the constants a , b and c such that 3

$$\sin^5 \theta = a \sin 5\theta + b \sin 3\theta + c \sin \theta.$$

(iii) Hence evaluate $16 \int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta$. 1

$$z^n = \text{cis}(n\theta)$$

$$\frac{1}{z^n} = \frac{1}{\cos n\theta + i \sin n\theta} \times \frac{\cos n\theta - i \sin n\theta}{\cos n\theta - i \sin n\theta}$$

$$\frac{1}{z^n} = \frac{\cos n\theta - i \sin n\theta}{\cos^2 \theta + \sin^2 \theta} = \cos n\theta - i \sin n\theta$$

$$z^n - \frac{1}{z^n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta = 2i \sin n\theta$$

Proofs involving De Moivre's are fine.

$$\left(z^1 - \frac{1}{z^1}\right)^5 = (2i \sin \theta)^5 = 2^5 i \sin^5 \theta$$

Now, expanding the LHS.

$$\begin{aligned} & \binom{5}{0} z^5 + \binom{5}{1} z^4 \times \left(-\frac{1}{z}\right)^1 + \binom{5}{2} z^3 \times \left(-\frac{1}{z}\right)^2 + \binom{5}{3} z^2 \times \left(-\frac{1}{z}\right)^3 + \binom{5}{4} z^1 \times \left(-\frac{1}{z}\right)^4 \\ & + \binom{5}{5} z^0 \times \left(-\frac{1}{z}\right)^5 \end{aligned}$$

$$z^5 - 5z^3 + 10z - 10\frac{1}{z} + 5\frac{1}{z^3} - \frac{1}{z^5} = \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$$

Then, using the identity from part (i) again.

$$\left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right) = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$$

Combining the two results,

$$2^5 i \sin^5 \theta = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$$

$$\sin^5 \theta = \frac{1}{2^4} \sin 5\theta - \frac{5}{2^4} \sin 3\theta + \frac{10}{2^4} \sin \theta$$

(iii) Hence evaluate $16 \int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta$.

1

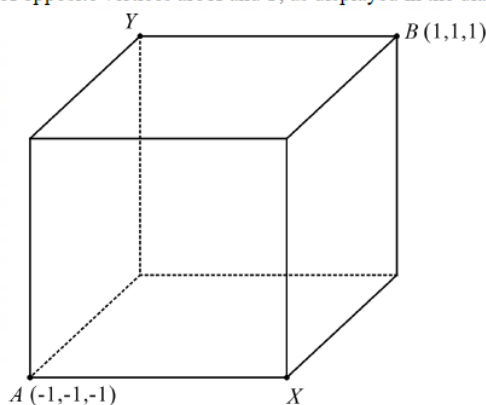
$$16 \int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta = 16 \int_0^{\frac{\pi}{2}} \left(\frac{1}{2^4} \sin 5\theta - \frac{5}{2^4} \sin 3\theta + \frac{10}{2^4} \sin \theta \right) d\theta$$

$$16 \int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta = \int_0^{\frac{\pi}{2}} (\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta) d\theta$$

$$16 \int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta = \left(-\frac{\cos 5\theta}{5} + \frac{5 \cos 3\theta}{3} - 10 \cos \theta \right)_0^{\frac{\pi}{2}}$$

$$16 \int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta = (0 - 0 + 0) - \left(-\frac{1}{5} + \frac{5}{3} - 10 \right) = \frac{128}{15}$$

- (d) A cube has a pair of opposite vertices $A (-1, -1, -1)$ and $B (1, 1, 1)$.
Another pair of opposite vertices are X and Y , as displayed in the diagram below.



If α is the acute angle between AB and XY , show that

$$\sin 2\alpha = \frac{2\sqrt{8}}{9}.$$

2

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$$

Without loss of generality we can assume that X and Y are formed to be parallel with the axes, so $X = (1, -1, -1)$ and $Y = (-1, 1, 1)$.

$$\overrightarrow{XY} = \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$$

$$|AB| = |XY| = \sqrt{12} = 2\sqrt{3}$$

$$\overrightarrow{AB} \cdot \overrightarrow{XY} = |\overrightarrow{AB}| \times |\overrightarrow{XY}| \cos \alpha = -4 + 4 + 4$$

$$12 \cos \alpha = 4$$

$$\cos \alpha = \frac{1}{3}$$

From an acute triangle or $(\sin^2 x = 1 - \cos^2 x)$ we can conclude that,

$$\sin \alpha = \frac{\sqrt{8}}{3}$$

$$\sin 2\alpha = 2 \cos \alpha \sin \alpha = 2 \times \frac{1}{3} \times \frac{\sqrt{8}}{3} = \frac{2\sqrt{8}}{9}$$

(e) Three positive number a , b and c satisfy the following conditions $a \geq b \geq c$ and $a + b + c \leq 1$.

By considering $(a + b + c)^2$ or otherwise, prove that $a^2 + 3b^2 + 5c^2 \leq 1$.

2

Initially, following the instructions.

$$a + b + c \leq 1$$

$$(a + b + c)^2 \leq 1$$

$$a^2 + ab + ac + b^2 + ab + bc + c^2 + ac + bc \leq 1$$

$$a^2 + b^2 + c^2 + 2ab + 2ac + 2bc \leq 1$$

Now, as

$$a \geq b \geq c$$

Then by multiplying by '2b'

$$2ab \geq 2b^2 \geq 2bc$$

Or multiplying by '2a'

$$2ac \geq 2bc \geq 2c^2$$

Which means,

$$2ab \geq 2b^2 \text{ ... and ... } 2ac \geq 2c^2 \text{ ... and ... } 2bc \geq c^2$$

$$a^2 + b^2 + c^2 + 2b^2 + 2c^2 + 2c^2 \leq a^2 + b^2 + c^2 + 2ab + 2ac + 2bc \leq 1$$

$$a^2 + 3b^2 + 5c^2 \leq 1$$

As required.

Question 14

(a) Use the substitution of $u = \sin x - x \cos x$, evaluate

3

$$\int_{\frac{\pi}{2}}^{\pi} \frac{x}{1 - x \cot x} dx.$$

$$u = \sin x - x \cos x$$

$$\frac{du}{dx} = \cos x - \cos x + x \sin x = x \sin x$$

$$\int \frac{x}{1 - x \cot x} dx = \int \frac{x}{1 - x \frac{\cos x}{\sin x}} dx = \int \frac{x \sin x}{\sin x - x \cos x} dx$$

$$\int \frac{x \sin x}{\sin x - x \cos x} dx = \int \frac{1}{u} du = \ln|u| + C = \ln|\sin x - x \cos x| + C$$

So...

$$\int_{\frac{\pi}{2}}^{\pi} \frac{x}{1 - x \cot x} dx = \left[\ln|\sin x - x \cos x| \right]_{\frac{\pi}{2}}^{\pi}$$

$$= \ln|0 - \pi \times -1| - \ln\left|1 - \frac{\pi}{2} \times 0\right| = \ln \pi$$

Obviously changing the bounds is a valid approach.

(b) Consider the following four equations, where $a > 0$ and $b > 0$.

$$D = \frac{2ab}{a+b}$$

$$U = \sqrt{ab}$$

$$B = \frac{a+b}{2}$$

$$O = \sqrt{\frac{a^2+b^2}{2}}$$

Show that $D \leq U \leq B \leq O$.

3

$$(\sqrt{a} - \sqrt{b})^2 \geq 0$$

$$a - 2\sqrt{ab} + b \geq 0$$

$$a + b \geq 2\sqrt{ab}$$

$$\frac{a+b}{2ab} \geq \frac{2\sqrt{ab}}{2ab}$$

$$\frac{a+b}{2ab} \geq \frac{1}{\sqrt{ab}}$$

$$\sqrt{ab} \geq \frac{2ab}{a+b}$$

$$U \geq D$$

From earlier,

$$a + b \geq 2\sqrt{ab}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$B \geq U$$

If you can't see a pathway, contradiction works! Assume $B > O$

$$\frac{a+b}{2} > \sqrt{\frac{a^2+b^2}{2}}$$

$$\frac{a^2 + 2ab + b^2}{4} > \frac{a^2 + b^2}{2}$$

$$0 > \frac{2a^2 + 2b^2 - a^2 - 2ab - b^2}{4}$$

$$0 > \frac{a^2 - 2ab + b^2}{4}$$

$$0 > \frac{(a-b)^2}{4}$$

Contradiction as $(a-b)^2 \geq 0$, therefore $B \leq O$ as required!

NOTE: Students should avoid assuming the AM-GM inequality in situations where it is naively similar.

- (c) The triangle ABC in the complex plane is formed from the points $A(\alpha)$, $B(\beta)$ and $C(\gamma)$. The complex numbers α , β and γ satisfy these two conditions:

$$\frac{\gamma - \alpha}{\beta - \alpha} = 1 - i \quad \text{and} \quad \alpha + \beta + \gamma = 0.$$

Calculate $|\alpha| : |\beta|$.

3

$$\gamma = -\alpha - \beta$$

$$\frac{\gamma - \alpha}{\beta - \alpha} = 1 - i$$

$$\frac{-2\alpha - \beta}{\beta - \alpha} = 1 - i$$

$$-2\alpha - \beta = \beta - i\beta - \alpha + i\alpha$$

$$-2\beta + i\beta = \alpha + i\alpha$$

$$\beta(-2 + i) = \alpha(1 + i)$$

$$\frac{\alpha}{\beta} = \frac{-2 + i}{1 + i}$$

$$\left| \frac{\alpha}{\beta} \right| = \left| \frac{-2 + i}{1 + i} \right|$$

$$\frac{|\alpha|}{|\beta|} = \frac{|-2 + i|}{|1 + i|}$$

$$\frac{|\alpha|}{|\beta|} = \frac{\sqrt{5}}{\sqrt{2}} = \frac{\sqrt{10}}{2}$$

Methods that involved geometric implications sadly got very muddled and confused – as the length of the numbers are not equal, so, yes, when considered like vectors they appear as at a triangle with a vertex at the origin. (But there isn't much more that can be concluded due to different magnitudes)

(d) Two magpies, A and B , are observed flying.

3

Their initial respective positions are:

$(\hat{i} - 2\hat{j} + 4\hat{k})$ metres and $(-2\hat{i} + a\hat{j} + 6\hat{k})$ metres, where a is a constant.

They are travelling with constant velocities and these are respectively:

$(2\hat{i} + 3\hat{j} + 6\hat{k})$ m/s and $(3\hat{i} + 12\hat{j} + 4\hat{k})$ m/s.

Determine the constant a , if the distance between the two magpies is least when $t = 5$ seconds.

$$v_A = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$$

$$x_A = \begin{pmatrix} 2t \\ 3t \\ 6t \end{pmatrix} + C$$

$$t = 0, x_A = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$$

$$x_A = \begin{pmatrix} 2t + 1 \\ 3t - 2 \\ 6t + 4 \end{pmatrix}$$

$$v_B = \begin{pmatrix} 3 \\ 12 \\ 4 \end{pmatrix}$$

$$x_B = \begin{pmatrix} 3t \\ 12t \\ 4t \end{pmatrix} + C$$

$$t = 0, x_B = \begin{pmatrix} -2 \\ a \\ 6 \end{pmatrix}$$

$$x_B = \begin{pmatrix} 3t - 2 \\ 12t + a \\ 4t + 6 \end{pmatrix}$$

Distance between the two,

$$\left| \begin{pmatrix} 3t - 2 \\ 12t + a \\ 4t + 6 \end{pmatrix} - \begin{pmatrix} 2t + 1 \\ 3t - 2 \\ 6t + 4 \end{pmatrix} \right| = \left| \begin{pmatrix} t - 3 \\ 9t + a + 2 \\ -2t + 2 \end{pmatrix} \right|$$

$$DIST = \sqrt{(t - 3)^2 + (9t + a + 2)^2 + (2 - 2t)^2}$$

When max distance occurs, so does max $DIST^2$

$$DIST^2 = t^2 - 6t + 9 + 81t^2 + 9at + 18t + 9at + a^2 + 2a + 18t + 2a + 4 + 4 - 8t + 4t^2$$

$$DIST^2 = 86t^2 + 22t + 18at + 17$$

This is a quadratic, so the minimum distance occurs at the turning point, when the derivative is zero.

$$\frac{d(DIST^2)}{dt} = 172t + 22 + 18a = 0$$

$t = 5$, is when it occurs (from question)

$$882 + 18a = 0$$

$$a = -49$$

NOTE: Using $t = 5$ before finding the max distance results in an incorrect result as time is a variable and the nature of the function in terms of time needs to be considered.

(e) By considering $(1+i)(4-i)^2$ prove that $\tan^{-1}\left(\frac{7}{23}\right) + 2 \tan^{-1}\left(\frac{1}{4}\right) = \frac{\pi}{4}$.

3

Expanding algebraically...

$$(1+i)(4-i)^2 = (1+i)(17-8i) = 23+7i$$

Then, considering the arguments

$$\text{Arg}(1+i) = \frac{\pi}{4}, \text{Arg}(4-i) = -\arctan\left(\frac{1}{4}\right), \text{Arg}(23+7i) = \arctan\left(\frac{7}{23}\right)$$

And so by applying the addition of arguments...

$$\frac{\pi}{4} + 2 \times -\arctan\left(\frac{1}{4}\right) = \arctan\left(\frac{7}{23}\right)$$

$$\frac{\pi}{4} = \arctan\left(\frac{7}{23}\right) + 2 \arctan\left(\frac{1}{4}\right)$$

Question 15

- (a) A body of **unit** mass falls vertically from rest, under gravity, from a height of 100 metres above the ground through a resisting medium. The resistance to its motion is $\frac{v^2}{100}$, where v is the speed of the body (in m/s) after falling a distance x metres.

(i) Show that the acceleration $\ddot{x}$ of the body is given by $\ddot{x} = g - \frac{v^2}{100}$. 1

(ii) Determine the terminal velocity V of the body. 1

(iii) Show that the velocity v of the body, after it has fallen a vertical distance x , is given by $v^2 = V^2 \left(1 - e^{-\frac{x}{50}} \right)$. 3

(iv) Determine the distance fallen (in metres) when the velocity of the body first reaches $\frac{V}{2}$. 2

$$F = ma$$

$$mg - \frac{v^2}{100} = ma$$

$$m = 1$$

$$a = mg - \frac{v^2}{100}$$

Terminal velocity occurs when acceleration is zero.

$$g - \frac{V^2}{100} = 0$$

$$V^2 = 100g$$

$$V = 10\sqrt{g}$$

$$v \times \frac{dv}{dx} = g - \frac{v^2}{100} = \frac{100g - v^2}{100}$$

$$\frac{v}{100g - v^2} dv = \frac{dx}{100}$$

$$\frac{-2v}{V^2 - v^2} dv = -\frac{dx}{50} \quad (\text{using } V^2 \text{ definition from earlier})$$

$$\ln|V^2 - v^2| = -\frac{x}{50} + C$$

$$t = 0, x = 0, v = 0$$

$$\ln|V^2| = C$$

$$\ln|V^2 - v^2| = -\frac{x}{50} + \ln(V^2)$$

$$V^2 - v^2 = e^{-\frac{x}{50}} \times e^{\ln(V^2)}$$

$$V^2 - v^2 = V^2 \times e^{-\frac{x}{50}}$$

$$v^2 = V^2 - V^2 \times e^{-\frac{x}{50}} = V^2 \left(1 - e^{-\frac{x}{50}}\right)$$

$$v = \frac{V}{2}, x = ?$$

$$\left(\frac{V}{2}\right)^2 = V^2 \left(1 - e^{-\frac{x}{50}}\right)$$

$$\frac{1}{4} = 1 - e^{-\frac{x}{50}}$$

$$e^{-\frac{x}{50}} = \frac{3}{4}$$

$$-\frac{x}{50} = \ln \frac{3}{4}$$

$$x = -50 \ln \frac{3}{4} = 50 \ln \frac{4}{3}$$

(b) Let $S_n = \sum_{p=1}^n \tan^{-1}\left(\frac{1}{2p^2}\right)$.

Prove by induction that $\tan(S_n) = \frac{n}{n+1}$.

4

Base case, $n = 1$

$$S_1 = \arctan\left(\frac{1}{2}\right)$$

$$\tan(S_1) = \frac{1}{2} = \frac{1}{1+1}$$

Assume, $n = k$

$$\tan(S_k) = \frac{k}{k+1}$$

Prove, $n = k + 1$

$$S_{k+1} = \arctan\left(\frac{1}{2(k+1)^2}\right) + S_k$$

$$\tan(S_{k+1}) = \tan\left(\arctan\left(\frac{1}{2(k+1)^2}\right) + S_k\right)$$

$$\tan(S_{k+1}) = \frac{\tan\left(\arctan\left(\frac{1}{2(k+1)^2}\right)\right) + \tan(S_k)}{1 - \tan\left(\arctan\left(\frac{1}{2(k+1)^2}\right)\right) \times \tan(S_k)}$$

$$\tan(S_{k+1}) = \frac{\frac{1}{2(k+1)^2} + \frac{k}{k+1}}{1 - \frac{1}{2(k+1)^2} \times \frac{k}{k+1}} \quad (\text{using the assumption and } \tan(\tan^{-1} x) = x)$$

Multiplying by $2(k+1)^3$

$$\tan(S_{k+1}) = \frac{(k+1) + 2k(k+1)^2}{2(k+1)^3 - k}$$

$$\tan(S_{k+1}) = \frac{(k+1)(1 + 2k(k+1))}{2k^3 + 6k^2 + 6k + 2 - k}$$

$$\tan(S_{k+1}) = \frac{(k+1)(2k^2 + 2k + 1)}{2k^3 + 6k^2 + 5k + 2}$$

Now, knowing that the desired denominator is $k+2$

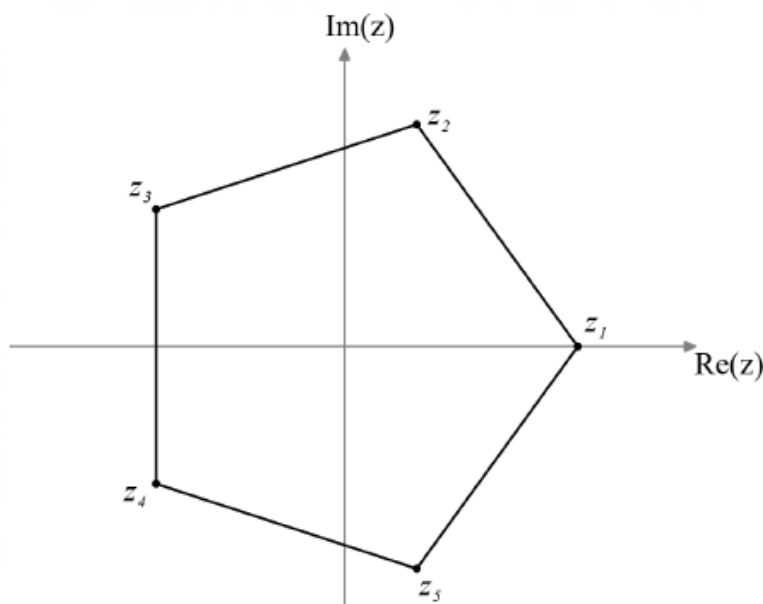
$$\tan(S_{k+1}) = \frac{(k+1)(2k^2 + 2k + 1)}{(k+2)(ak^2 + bk + c)}$$

Applying polynomial division / by observation

$$\tan(S_{k+1}) = \frac{(k+1)(2k^2 + 2k + 1)}{(k+2)(2k^2 + 2k + 1)} = \frac{k+1}{k+2}$$

As required!

- (c) The solutions of $z^5 = 1$ can be joined to form a regular pentagon in the complex plane, as shown below.



- (i) For $n \geq 3$, show that the perimeter of the regular n -gon formed by the solutions of $z^n = 1$ is $P_n = 2n \sin\left(\frac{\pi}{n}\right)$.

2

- (ii) It is known that as $x \rightarrow 0$ then $\frac{\sin x}{x} \rightarrow 1$. (Do NOT prove this.)
Using part (i), show that as $n \rightarrow \infty$, $P_n \rightarrow 2\pi$.

2

The length of one side, S , can be found with the cosine rule.

The angle is $\frac{2\pi}{n}$ and the adjacent sides are 1.

$$S^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \times \cos\left(\frac{2\pi}{n}\right)$$

$$S^2 = 2 \left(1 - \cos\left(2 \times \frac{\pi}{n}\right)\right)$$

From the reference sheet,

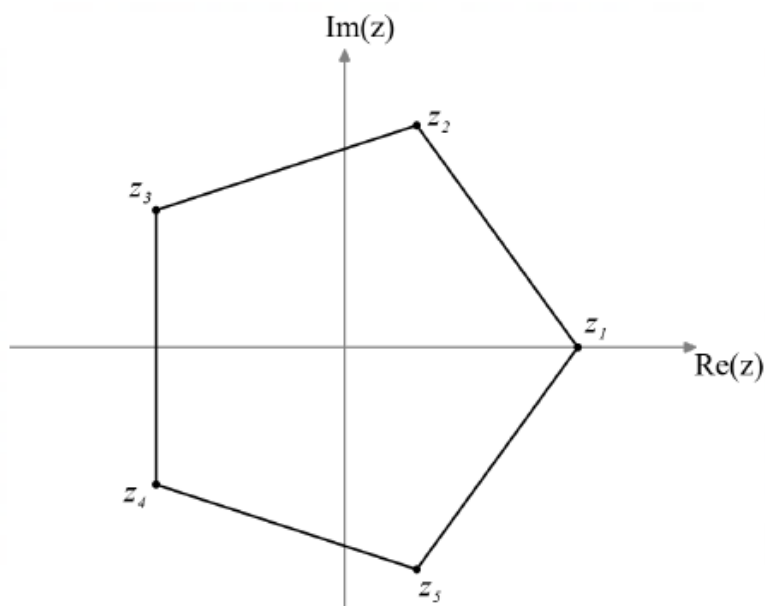
$$S^2 = 2 \times 2 \sin^2 \frac{\pi}{n}$$

$$S = 2 \sin \frac{\pi}{n}$$

There are n sides, so

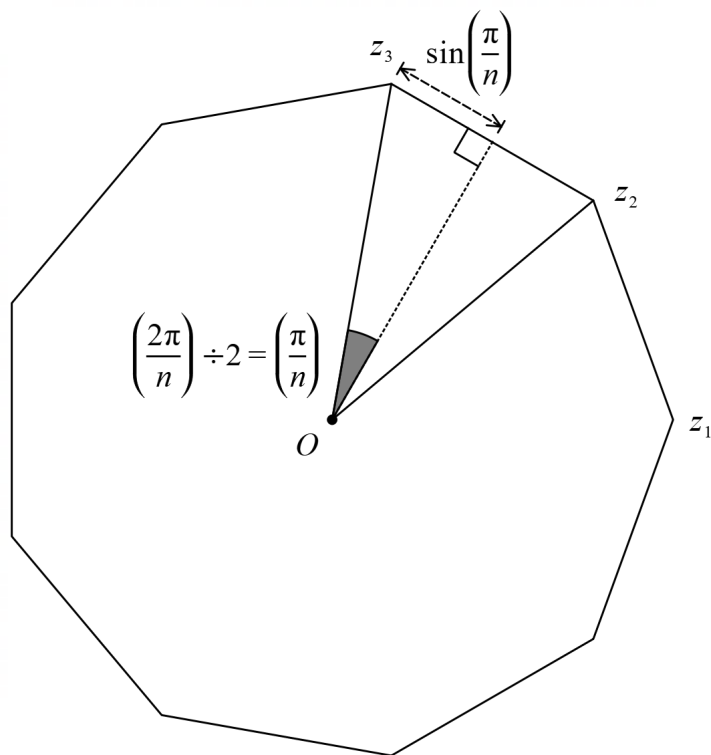
$$P_n = 2n \sin \frac{\pi}{n}$$

- (c) The solutions of $z^5 = 1$ can be joined to form a regular pentagon in the complex plane, as shown below.



- (i) For $n \geq 3$, show that the perimeter of the regular n -gon formed by the solutions of $z^n = 1$ is $P_n = 2n \sin\left(\frac{\pi}{n}\right)$.

2



Alternatively, you could use right angled trig to find the length of half of one of those sides.

As, $|z_n| = 1$, then

$$\sin\left(\frac{\pi}{n}\right) = \frac{\text{length}}{1}$$

This then concludes that the length of one side is $2 \sin\left(\frac{\pi}{n}\right)$

And then $P_n = 2n \sin\left(\frac{\pi}{n}\right)$

(ii) It is known that as $x \rightarrow 0$ then $\frac{\sin x}{x} \rightarrow 1$. (Do NOT prove this.)

Using part (i), show that as $n \rightarrow \infty$, $P_n \rightarrow 2\pi$.

2

As $n \rightarrow \infty$

$$\frac{\pi}{n} \rightarrow 0$$

So,

$$\frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \rightarrow 1$$

$$\frac{n \sin\left(\frac{\pi}{n}\right)}{\pi} \rightarrow 1$$

$$n \sin\left(\frac{\pi}{n}\right) \rightarrow \pi$$

$$2n \sin\left(\frac{\pi}{n}\right) \rightarrow 2\pi$$

As required!

Question 16

(a) Let $I_n = \int_0^{\frac{1}{4}} \frac{x^n}{e^x} dx$ where $n = 1, 2, 3, \dots$

(i) Calculate I_1 .

2

$$I_1 = \int_0^{\frac{1}{4}} \frac{x}{e^x} dx = \int_0^{\frac{1}{4}} x e^{-x} dx$$

Integration by parts,

$$u = x$$

$$v = -e^{-x}$$

$$u' = 1$$

$$v' = e^{-x}$$

$$I_1 = \left(-x e^{-x} \right)_0^{\frac{1}{4}} - \int_0^{\frac{1}{4}} 1 \times -e^{-x} dx$$

$$I_1 = -\frac{1}{4} \times e^{-\frac{1}{4}} - 0 - \left(e^{-x} \right)_0^{\frac{1}{4}}$$

$$I_1 = -\frac{1}{4e^{\frac{1}{4}}} - \left(e^{-\frac{1}{4}} - e^0 \right)$$

$$I_1 = 1 - \frac{1}{e^{\frac{1}{4}}} - \frac{1}{4e^{\frac{1}{4}}} = 1 - \frac{5}{4e^{\frac{1}{4}}}$$

(ii) Show that $kI_k = I_{k+1} - I_k + \frac{\left(\frac{1}{4}\right)^{k+1}}{e^{\frac{1}{4}}}$.

3

$$I_{k+1} = \int_0^{\frac{1}{4}} \frac{x^{k+1}}{e^x} dx$$

By parts,

$$u = x^{k+1}$$

$$v = -e^{-x}$$

$$u' = (k+1)x^k$$

$$v' = e^{-x}$$

$$I_{k+1} = \left(-x^{k+1}e^{-x}\right)_0^{\frac{1}{4}} - \int_0^{\frac{1}{4}} (k+1)x^k \times -e^{-x} dx$$

$$I_{k+1} = -\left(\frac{1}{4}\right)^{k+1} e^{-\frac{1}{4}} - 0 + (k+1) \int_0^{\frac{1}{4}} \frac{x^k}{e^x} dx$$

$$I_{k+1} = \frac{-\left(\frac{1}{4}\right)^{k+1}}{e^{\frac{1}{4}}} + (k+1)I_k$$

$$I_{k+1} - I_k + \frac{\left(\frac{1}{4}\right)^{k+1}}{e^{\frac{1}{4}}} = kI_k$$

As required.

(Alternative approach on next page.)

(ii) Show that $kI_k = I_{k+1} - I_k + \frac{\left(\frac{1}{4}\right)^{k+1}}{e^{\frac{1}{4}}}$.

3

$$I_k = \int_0^{\frac{1}{4}} \frac{x^k}{e^x} dx$$

By parts,

$$u = e^{-x}$$

$$v = \frac{x^{k+1}}{k+1}$$

$$u' = -e^{-x}$$

$$v' = x^k$$

$$I_k = \left(e^{-x} \times \frac{x^{k+1}}{k+1} \right)_0^{\frac{1}{4}} - \int_0^{\frac{1}{4}} (-e^{-x}) \times \left(\frac{x^{k+1}}{k+1} \right) dx$$

$$I_k = \left[\frac{\left(\frac{1}{4}\right)^{k+1} e^{-\frac{1}{4}}}{k+1} - 0 \right] + \frac{1}{k+1} \int_0^{\frac{1}{4}} \frac{x^{k+1}}{e^x} dx$$

$$I_k = \frac{\left(\frac{1}{4}\right)^{k+1} e^{-\frac{1}{4}}}{k+1} + \frac{I_{k+1}}{k+1}$$

$$(k+1)I_k = \frac{\left(\frac{1}{4}\right)^{k+1}}{e^{\frac{1}{4}}} + I_{k+1}$$

$$kI_k = I_{k+1} - I_k + \frac{\left(\frac{1}{4}\right)^{k+1}}{e^{\frac{1}{4}}}$$

As required.

NOTE, if you accidentally get an expression involving 'k-1' – you can substitute $u = k - 1$, $u = k$, and $u + 1 = k + 1$

This should then get you to the desired result (if you are careful...)

(iii) Deduce that $\sum_{k=1}^n kI_k = I_{n+1} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^n}{12e^{\frac{1}{4}}}$.

3

Induction: $n = 1$ From part (ii)

$$\sum_{k=1}^1 kI_k = I_1 = I_2 - I_1 + \frac{\left(\frac{1}{4}\right)^2}{e^{\frac{1}{4}}} = I_2 - I_1 + \frac{\frac{1}{16}}{e^{\frac{1}{4}}} = I_2 - I_1 + \frac{1}{16e^{\frac{1}{4}}}$$

Desired result is,

$$\sum_{k=1}^1 kI_k = I_2 - I_1 + \frac{1 - \left(\frac{1}{4}\right)^1}{12e^{\frac{1}{4}}} = I_2 - I_1 + \frac{1}{16e^{\frac{1}{4}}}$$

As required.

Assume: $n = Q$

$$\sum_{k=1}^Q kI_k = I_{Q+1} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^Q}{12e^{\frac{1}{4}}}$$

Prove: $n = Q + 1$

$$\begin{aligned} \sum_{k=1}^{Q+1} kI_k &= \sum_{k=1}^Q kI_k + (Q+1)I_{Q+1} \\ &= I_{Q+1} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^Q}{12e^{\frac{1}{4}}} + (Q+1)I_{Q+1} \text{ (Assumption)} \end{aligned}$$

From part (ii)

$$\begin{aligned} (Q+1)I_{Q+1} &= I_{Q+2} - I_{Q+1} + \frac{\left(\frac{1}{4}\right)^{Q+2}}{e^{\frac{1}{4}}} \\ \sum_{k=1}^{Q+1} kI_k &= I_{Q+1} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^Q}{12e^{\frac{1}{4}}} + I_{Q+2} - I_{Q+1} + \frac{\left(\frac{1}{4}\right)^{Q+2}}{e^{\frac{1}{4}}} \\ \sum_{k=1}^{Q+1} kI_k &= I_{Q+2} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^Q + 12\left(\frac{1}{4}\right)^{Q+2}}{12e^{\frac{1}{4}}} \end{aligned}$$

$$\sum_{k=1}^{Q+1} kI_k = I_{Q+2} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^Q \left(1 - 12\left(\frac{1}{4}\right)^2\right)}{12e^{\frac{1}{4}}} = I_{Q+2} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^{Q+1}}{12e^{\frac{1}{4}}}$$

As required!

Alternatively, the sum is in fact telescoping – that is.

$$\sum_{k=1}^n kI_k = 1I_1 + 2I_2 + 3I_3 + \cdots + (n-1)I_{n-1} + nI_n$$

Using part (ii)

$$\sum_{k=1}^n kI_k = \left(I_2 - I_1 + \frac{\left(\frac{1}{4}\right)^2}{e^{\frac{1}{4}}} \right) + \left(I_3 - I_2 + \frac{\left(\frac{1}{4}\right)^3}{e^{\frac{1}{4}}} \right) + \left(I_4 - I_3 + \frac{\left(\frac{1}{4}\right)^4}{e^{\frac{1}{4}}} \right) + \cdots + \left(I_n - I_{n-1} + \frac{\left(\frac{1}{4}\right)^n}{e^{\frac{1}{4}}} \right) + \left(I_{n+1} - I_n + \frac{\left(\frac{1}{4}\right)^{n+1}}{e^{\frac{1}{4}}} \right)$$

So, as this sum keeps going you see that the middle I_k terms cancel out! So, we are left with.

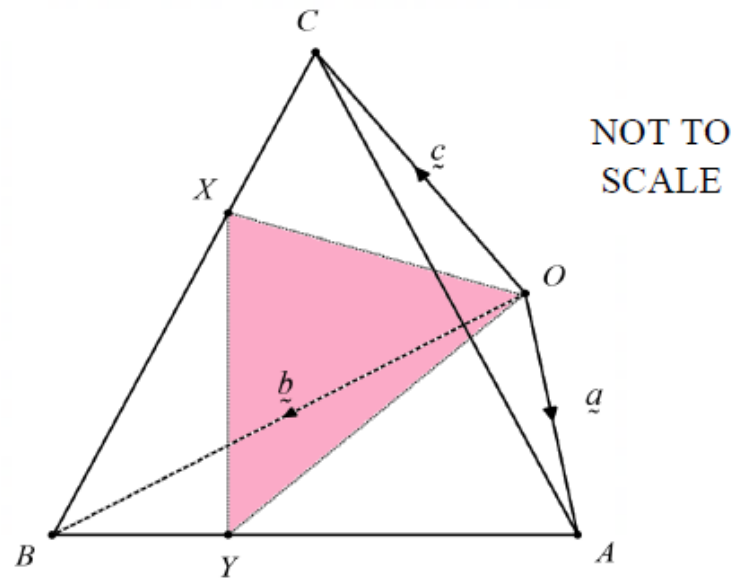
$$\sum_{k=1}^n kI_k = -I_1 + I_{n+1} + \frac{1}{e^{\frac{1}{4}}} \left[\left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \cdots + \left(\frac{1}{4}\right)^{n+1} \right]$$

That last term is a GP with n-terms and $a = \left(\frac{1}{4}\right)^2$, $r = \left(\frac{1}{4}\right)$

$$\sum_{k=1}^n kI_k = -I_1 + I_{n+1} + \frac{1}{e^{\frac{1}{4}}} \times \left[\frac{\left(\frac{1}{4}\right)^2 \left(1 - \left(\frac{1}{4}\right)^n\right)}{1 - \frac{1}{4}} \right] = I_{n+1} - I_1 + \frac{1 - \left(\frac{1}{4}\right)^n}{12e^{\frac{1}{4}}}$$

As required!

(b) The diagram shows a regular tetrahedron $OABC$ with sides of length 1.



Let X and Y be points on the edges such that $|\overrightarrow{BX}| : |\overrightarrow{XC}| = |\overrightarrow{AY}| : |\overrightarrow{YB}| = x : 1 - x$, where $0 < x < 1$.

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, and $\overrightarrow{OC} = \underline{c}$.

(i) Show that $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{c} = \frac{1}{2}$.

1

**As $OABC$ is a regular tetrahedron, each side is an equilateral triangle.
Hence, the angles formed are $\frac{\pi}{3}$ radians.**

$$\underline{a} \cdot \underline{b} = |\underline{a}| \times |\underline{b}| \times \cos \frac{\pi}{3} = \frac{1}{2}$$

Similarly,

$$\underline{a} \cdot \underline{c} = |\underline{a}| \times |\underline{c}| \times \cos \frac{\pi}{3} = \frac{1}{2}$$

$$\underline{b} \cdot \underline{c} = |\underline{b}| \times |\underline{c}| \times \cos \frac{\pi}{3} = \frac{1}{2}$$

(ii) Show that $|\overrightarrow{OY}| = \sqrt{1-x+x^2}$.

2

$$\overrightarrow{OY} = \overrightarrow{OB} + \overrightarrow{BY}$$

$$\overrightarrow{OY} = \mathbf{b} + (1-x)\overrightarrow{BA}$$

$$\overrightarrow{OY} = \mathbf{b} + (1-x)(-\mathbf{b} + \mathbf{a})$$

$$\overrightarrow{OY} = (1-1+x)\mathbf{b} + (1-x)\mathbf{a}$$

$$\overrightarrow{OY} = (1-x)\mathbf{a} + (x)\mathbf{b}$$

$$|\overrightarrow{OY}|^2 = \overrightarrow{OY} \cdot \overrightarrow{OY} = [(1-x)\mathbf{a} + (x)\mathbf{b}] \cdot [(1-x)\mathbf{a} + (x)\mathbf{b}]$$

$$|\overrightarrow{OY}|^2 = (1-x)^2 \times \mathbf{a} \cdot \mathbf{a} + 2x(1-x) \times \mathbf{a} \cdot \mathbf{b} + x^2 \times \mathbf{b} \cdot \mathbf{b}$$

From part (i) and that the lengths are 1.

$$|\overrightarrow{OY}|^2 = (1-x)^2 \times 1 + 2x(1-x) \times \frac{1}{2} + x^2 \times 1$$

$$|\overrightarrow{OY}|^2 = 1 - 2x + x^2 + x - x^2 + x^2$$

$$|\overrightarrow{OY}|^2 = 1 - x + x^2$$

$$|\overrightarrow{OY}| = \sqrt{1-x+x^2}$$

NOTE: Length needs to be found using the dot-product as 'a' and 'b' are not perpendicular to each other, so you cannot use the formula on the reference sheet.

$$\overrightarrow{OX} \cdot \overrightarrow{OY} = |\overrightarrow{OX}| \times |\overrightarrow{OY}| \times \cos \angle YOX$$

So, we've found $\overrightarrow{OY}$ earlier, we need to do the same $\overrightarrow{OX}$.

$$\overrightarrow{OX} = \overrightarrow{OC} + \overrightarrow{CX}$$

$$\overrightarrow{OX} = c + (1-x)\overrightarrow{CB}$$

$$\overrightarrow{OX} = c + (1-x)(-c + b)$$

$$\overrightarrow{OX} = (x)c + (1-x)b$$

$$\overrightarrow{OX} = (1-x)b + (x)c$$

This is identical to $\overrightarrow{OY}$, so we can conclude that

$$|\overrightarrow{OX}| = \sqrt{1-x+x^2}$$

Going back to the dot-product.

$$\overrightarrow{OX} \cdot \overrightarrow{OY} = |\overrightarrow{OX}| \times |\overrightarrow{OY}| \times \cos \angle YOX$$

$$\{(1-x)b + (x)c\} \cdot \{(1-x)a + (x)b\} = \sqrt{1-x+x^2} \times \sqrt{1-x+x^2} \times \cos \angle YOX$$

$$\cos \angle YOX = \frac{(1-x)^2 \times a \cdot b + (1-x)x \times b \cdot b + x(1-x) \times a \cdot c + x^2 \times b \cdot c}{1-x+x^2}$$

As $a \cdot b = a \cdot c = b \cdot c = \frac{1}{2}$ and $b \cdot b = 1$, **factorizing out** $\frac{1}{2}$

$$\cos \angle YOX = \frac{(1-x)^2 + 2x(1-x) + x(1-x) + x^2}{2(1-x+x^2)}$$

$$\cos \angle YOX = \frac{1-2x+x^2+2x-2x^2+x-x^2+x^2}{2(1-x+x^2)} = \frac{1+x-x^2}{2(1-x+x^2)}$$

$$\cos \angle YOX = \frac{1 + x - x^2}{2(1 - x + x^2)}$$

The range of values of cosine will determine the range of possible values of the angle.

As $0 < x < 1$ – the max/min will occur at either the ends and/or when the derivative is zero.

$$x = 1, \text{ and } x = 0, \cos \angle YOX = \frac{1}{2}$$

$$\angle YOX = \frac{\pi}{3}$$

Now, for the derivative...

$$\frac{d(\cos \angle YOX)}{dx} = \frac{1}{2} \times \frac{(1 - x + x^2) \times (1 - 2x) - (1 + x - x^2) \times (-1 + 2x)}{(1 - x + x^2)^2}$$

$$\frac{d(\cos \angle YOX)}{dx} = \frac{1}{2} \times \frac{1 - 2x - x - 2x^2 + x^2 - 2x^3 + 1 - 2x + x - 2x^2 - x^2 + 2x^3}{(1 - x + x^2)^2}$$

$$\frac{d(\cos \angle YOX)}{dx} = \frac{1}{2} \times \frac{2 - 4x}{(1 - x + x^2)^2}$$

When this is zero,

$$0 = \frac{1 - 2x}{(1 - x + x^2)^2}$$

$$x = \frac{1}{2}$$

$$x = 0, \frac{d(\cos \angle YOX)}{dx} > 0$$

$$x = 1, \frac{d(\cos \angle YOX)}{dx} < 0$$

Hence, $x = \frac{1}{2}$ is a maximum.

So,

$$\cos \angle YOX = \frac{1 + \left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)^2}{2\left(1 - \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2\right)} = \frac{5}{6}$$

Conclusion is that,

$$\frac{1}{2} < \cos \angle YOX \leq \frac{5}{6}$$

Note, it is strictly greater than $\frac{1}{2}$ as $x \neq 0$ or 1 .

$$\arccos\left(\frac{5}{6}\right) \leq \angle YOX \leq \frac{\pi}{3}$$

Note, the change in direction as cosine is decreasing in $\left[0, \frac{\pi}{2}\right]$.

ALTERNATIVE method to finding the max:

$$\cos \angle YOX = \frac{1 + x - x^2}{2(1 - x + x^2)}$$

Start completing the square...

$$\cos \angle YOX = \frac{-(x^2 - x) + 1}{2(x^2 - x + 1)} = \frac{-(x^2 - x + \frac{1}{4} - \frac{1}{4}) + 1}{2(x^2 - x + \frac{1}{4} - \frac{1}{4} + 1)}$$

$$= \frac{-\left[\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}\right] + 1}{2\left[\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}\right]} = \frac{1}{2} \times \frac{\frac{5}{4} - \left(x - \frac{1}{2}\right)^2}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}}$$

Maximum occurs when denominator is smallest and numerator is largest. Clearly, this is when $x = \frac{1}{2}$ as

$$\frac{5}{4} - \left(x - \frac{1}{2}\right)^2 \leq \frac{5}{4} \text{ and } \left(x - \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$$